

Wind Farm Project Finance Case Study

Commercial bank management, capital structure, cash-flow waterfall, and covenant analysis

Executive Summary

This project evaluates the financing feasibility of a hypothetical \$500 million wind power facility in South Dakota. The model combines project-finance capital structure, operating cash-flow projections, debt-service scheduling, reserve funding, and lender-style covenant analysis.

The final model used senior debt, mezzanine capital, sponsor equity, and limited-recourse support to preserve project feasibility while meeting the bank covenant package. Stress cases tested whether the project could still satisfy return and coverage targets under lower electricity generation and lower power prices.

Key Model Inputs

Input	Base-case assumption
Capital need	\$500 million
Year 2 production	7.0 million MWh
Years 3-9 production	10.0 million MWh annually
Power price	\$15/MWh in year 2; \$18/MWh afterward
Senior debt	\$325 million at 5%
Mezzanine capital	\$125 million at 14%
Initial sponsor equity	\$50 million
Additional sponsor support	\$72.5 million in the base model
Tax rate	21%

Financing Structure

Wind Farm Project Finance: Capital Structure

\$500 million capital need financed with senior debt, mezzanine capital, and sponsor equity/support.

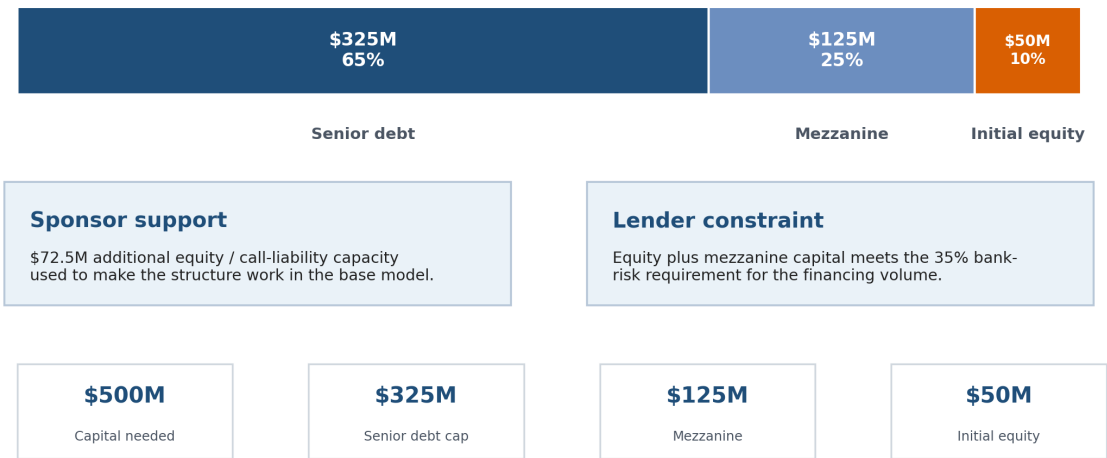


Figure: Capital stack and sponsor-support logic used in the FIN-416 wind farm project finance model.

Figure: Capital stack and sponsor-support logic used in the project finance model.

Scenario Results

The project was tested under a base case and two stress cases. The core feasibility question was whether the model could preserve sponsor return while satisfying the bank covenant package.

Scenario	Operating stress	ROE	Min ADSCR	Min LLCR	Min ICR	Covenants
Part A - base	Base generation and price	29.1%	1.60+	2.31+	1.76+	Met
Part B - volume stress	3% MWh decline	25.2%	1.60+	2.30+	1.76+	Met
Part C - volume + price stress	2% MWh decline; 3% price decline in years 6-9	25.0%	1.60+	2.28+	1.76+	Met

Interpretation

The base case produced a return above the required 25% threshold. The two downside cases remained near or above the target return while preserving the required covenant thresholds, which is the central lender-facing feasibility signal in the model.

Cash-Flow and Debt-Service Review

Wind Farm Cash-Flow and Debt-Service Summary

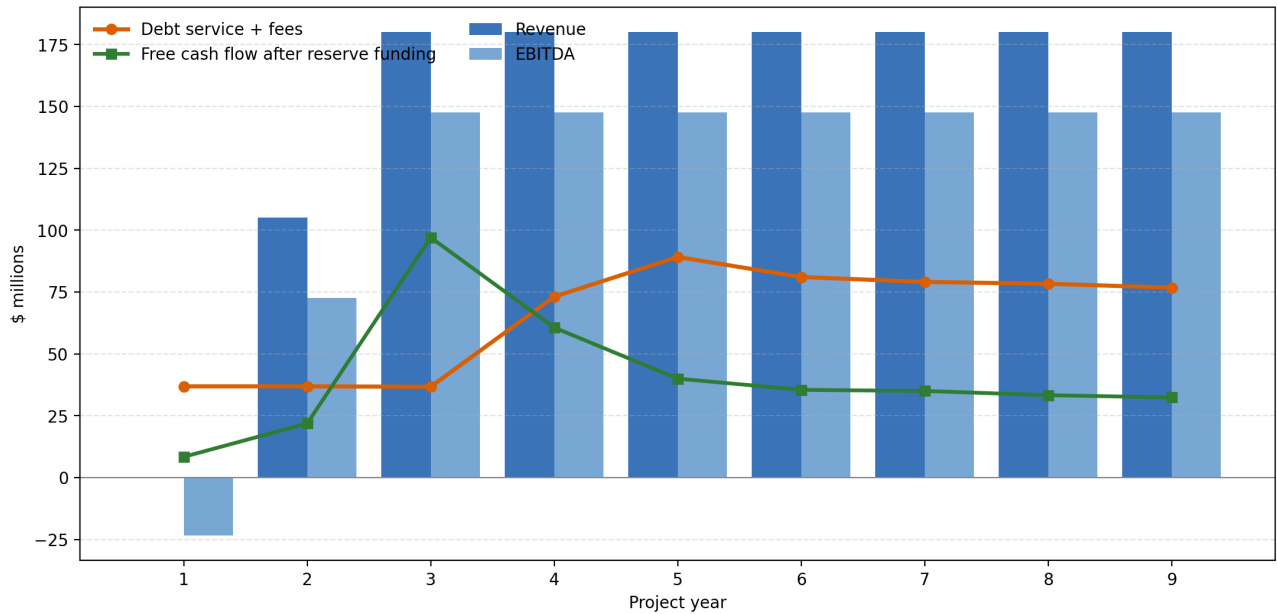


Figure: Base-case operating cash flow, debt-service burden, and free cash flow in the project finance model.

Figure: Revenue, EBITDA, debt-service burden, and free cash flow in the base model.

The cash-flow pattern shows why project finance relies on timing as well as profitability: the model has to support debt service, reserve funding, and eventual sponsor distributions after construction and ramp-up.

Credit Analysis Framework

Wind Farm Credit Analysis Framework

From operating assumptions to lender-style feasibility and covenant review.

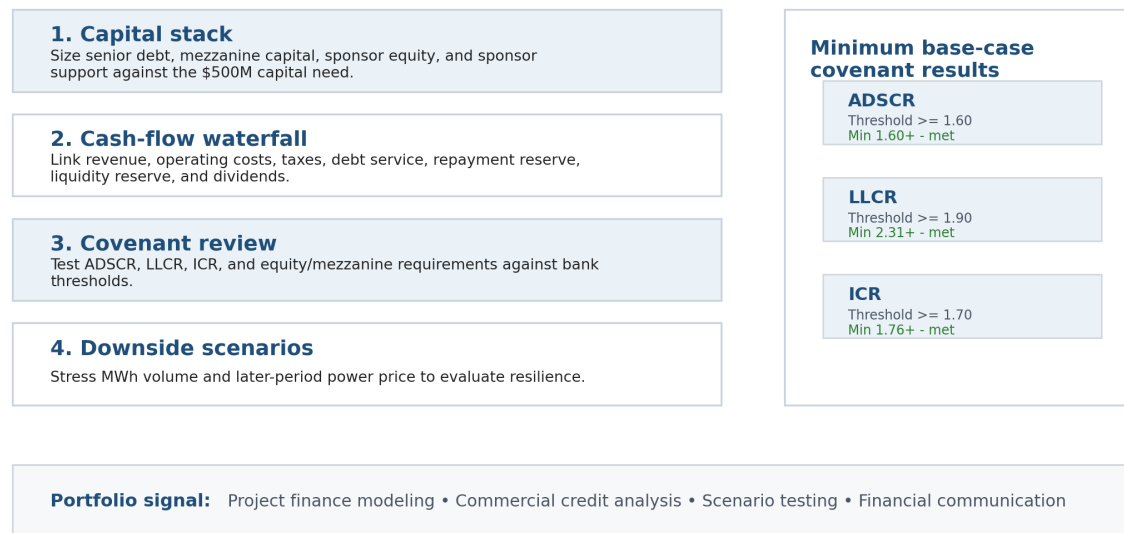


Figure: Framework for converting a project-finance model into a lender-style credit assessment.

Figure: Framework for translating the model into a lender-style credit assessment.

Portfolio Relevance

This project demonstrates lender-style project finance thinking. It shows how a finance model moves from project assumptions to capital structure, operating cash flow, debt-service coverage, liquidity reserves, and downside sensitivity analysis.

- Project-finance capital structure
- Cash-flow waterfall construction
- Senior debt and mezzanine repayment logic
- Debt-service coverage analysis
- Loan-life cover ratio and interest cover ratio review
- Scenario and sensitivity analysis
- Commercial credit interpretation
- Excel-based financial modeling

Caution

This portfolio package is based on a classroom project finance case. It is intended to demonstrate modeling logic, credit-analysis reasoning, and communication of results rather than serve as an investment recommendation or legal credit opinion.